

Computer Programming - C

Lab Assignment 1

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1. //Write a C program to add, subtract, multiply, and divide two numbers.

```
#include<stdio.h>
int main() {
    int num1, num2;

    printf("Enter two numbers: ");
    scanf("%d %d", &num1, &num2);

    printf("Sum: %d\n", num1 + num2);
    printf("Difference: %d\n", num1 - num2);
    printf("Product: %d\n", num1 * num2);
    printf("Quotient: %d\n", num1 / num2);

    return 0;
}
```

2. //WAP to input a fraction (rational number) and display. (Ask the user to input numerator
//and denominator, then display it in the form of p/q without simplification)

```
#include<stdio.h>
int main() {
    int num1, num2;
    printf("Enter numerator: ");
    scanf("%d", &num1);
    printf("Enter denominator: ");
    scanf("%d", &num2);
    printf("Fraction: %d/%d\n", num1, num2);
    return 0;
}
```

3. //WAP to Print the ASCII Value of a Character.

```
#include<stdio.h>
int main() {
    char ch;
    printf("Enter a character: ");
    scanf("%c", &ch);
    printf("ASCII value: %d\n", ch);
    return 0;
}
```

4. // WAP to Swap Two Numbers.

```
#include<stdio.h>
int main() {
    int a, b, temp;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Before swapping: a = %d, b = %d\n", a, b);

    temp = a;
    a = b;
    b = temp;
    printf("After swapping: a = %d, b = %d\n", a, b);
    return 0;
}
```

5. //Write a program to find the area of a circle.

```
#include<stdio.h>
int main(){
    float radius, area;
    printf("Enter the radius of the circle: ");
    scanf("%f", &radius);
    area = 3.14 * radius * radius;
    printf("Area of the circle: %.2f\n", area);
    return 0;
}
```

6. // WAP to calculate simple interest.

```
#include <stdio.h>
int main() {
    float principal, rate, time, interest;
    printf("Enter the principal amount: ");
    scanf("%f", &principal);
    printf("Enter the rate of interest: ");
    scanf("%f", &rate);
    printf("Enter the time period: ");
    scanf("%f", &time);
    interest = principal * rate * time;
    printf("Simple interest: %.2f\n", interest); //%.2f format specifier for floating value upto
2 decimal places.
    return 0;
}
```

7. //Writa a program to calculate compound interest.
// Compounded Annually.
// Time period in years.

```
#include <stdio.h>
int main() {
    float principal, rate, time, interest;
    printf("Enter the principal amount: ");
    scanf("%f", &principal);
    printf("Enter the rate of interest: ");
    scanf("%f", &rate);
    printf("Enter the time period: ");
    scanf("%f", &time);
    interest = principal * (1 + rate) * time;
    printf("Compound interest: %.2f\n", interest);
    return 0;
}
```

8. /* WAP to calculate gross salary of a person, where gross_salary=basic+da+ta and da is 10% of basic and ta is 12% of basic. */

```
#include <stdio.h>
int main() {
    float basic, da, ta, gross_salary;
    printf("Enter basic salary: ");
    scanf("%f", &basic);
    da = 0.1 * basic;
    ta = 0.12 * basic;
    gross_salary = basic + da + ta;
    printf("Gross salary: %.2f\n", gross_salary);
    return 0;
}
```

9. // WAP to Convert Fahrenheit to Celcius in C.

```
#include<stdio.h>
int main() {
    float fahrenheit, celcius;
    printf("Enter temperature in Fahrenheit: ");
    scanf("%f", &fahrenheit);
    celcius = (fahrenheit - 32) * 5 / 9;
    printf("Temperature in Celcius: %.2f\n", celcius);
    return 0;
}
```

10. /* You plan a trip to Europe costing ₹1,50,000. You take a travel loan from a bank at a simple interest rate of 8% per annum for 2 years. Write a C program to calculate the total interest you will pay and the total amount to be repaid at the end of the term. */

```
#include <stdio.h>
int main() {
    float principal = 150000;
    float rate = 0.08;
    int time = 2;
    float interest = principal * rate * time;
    float total = principal + interest;
    printf("Total interest: %.2f\n", interest);
    printf("Total amount: %.2f\n", total);
    return 0;
}
```

11. /* You have a circular garden in your backyard with a radius of 7.5 meters. You want to cover the garden with grass turf, and the turf is sold at ₹55 per square meter. Write a C program to calculate the area of the garden and the total cost of covering it with turf. */

```
#include <stdio.h>
int main() {
    float radius = 7.5;
    float area = 3.14 * radius * radius;
    float cost = area * 55;
    printf("Area: %.2f\n", area);
    printf("Cost: %.2f\n", cost);
    return 0;
}
```

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